

Asset Maintenance History and Inspection Records Analysis



[UC58] - Awesense Platform Use Case

The Utility Problem

Every distribution asset accumulates a history over its operational life — a record of maintenance interventions, inspection findings, outage involvements, and condition assessments. This history is one of the most valuable sources of insight available to an asset manager: an asset that has required repeated corrective maintenance, appeared in multiple outage records, or shown a consistent pattern of degrading inspection results is telling a story about its reliability trajectory. Acting on that story early — before the next failure — is the core promise of condition-based and predictive asset management.

In practice, however, this history is rarely accessible in a coherent, asset-centric form. Maintenance records live in the work order management system. Outage involvement is recorded in the OMS. Inspection results may exist in a separate inspection management system, a shared drive of PDFs, or paper files in a district office. These systems are not integrated, and none of them provides a unified, chronological view of everything that has happened to a specific asset over its lifetime. When an engineer wants to investigate a suspect transformer or cable section, they must manually pull records from multiple systems, cross-reference asset identifiers that may not be consistent across those systems, and assemble the history by hand — a process that can take hours for a single asset.

The inability to efficiently query asset history also means that recurring issues go unrecognized at scale. A fuse that has been replaced three times in five years, a cable section that appears in outage records every winter, or a transformer that has failed inspection on consecutive cycles — these patterns exist in the data but are invisible without a system capable of filtering and aggregating maintenance and inspection records across the full fleet. Without that capability, recurring issues are addressed reactively, one event at a time, rather than being identified as systemic problems deserving a coordinated intervention.

The Data Solutions and Methods

The Awesense Platform integrates data from the utility's work order management system, OMS, and GIS into the Awesense Energy Data Model (EDM) — a unified, asset-centric data layer in which maintenance records, outage events, and inspection results are all linked to the same asset identifier used in the GIS. This integration resolves the cross-system identifier inconsistency problem and creates, for the first time, a single queryable source of the full operational history for every asset in the fleet. Unstructured documents such as inspection reports and condition assessments can additionally be attached directly to the corresponding GIS asset object within the EDM, co-locating structured and unstructured history in the same place.

The platform exposes this unified asset history through its SQL API and REST API. External analytical workflows — for example, Python-based pipelines — can query the EDM to retrieve the

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complete maintenance and inspection record for any asset or group of assets, with filtering across several dimensions. By outage type, a user can isolate assets that have appeared in sustained interruptions versus momentary events, or focus on specific cause codes such as equipment failure, tree contact, or animal contact. By time domain, queries can be scoped to a specific year, season, or rolling window — enabling seasonal pattern detection and year-over-year comparisons. By location, records can be filtered to a feeder, substation zone, geographic polygon, or administrative boundary such as a municipal area or rate zone.

With this filtered, asset-centric history in hand, recurring issue detection becomes tractable. For each asset, the analytical pipeline computes summary statistics across the historical record: total corrective work orders, count of outage involvements, trend in inspection condition ratings over successive cycles, and mean time between maintenance events. Assets that exceed configurable thresholds on any of these dimensions — for example, three or more corrective work orders within a five-year window, or two consecutive failing inspection ratings — are surfaced as candidates for escalated attention. The output is an asset-level ranking that prioritizes chronic underperformers for engineering review, enabling maintenance and capital planning teams to shift from reactive repair to proactive, evidence-based intervention.

Results can be visualized geospatially in True Grid Intelligence (TGI), the web-based digital twin explorer of the Awesense Platform, allowing teams to see the geographic concentration of high-maintenance assets and identify whether recurring issues cluster on specific feeders, in specific geographic areas, or around particular equipment vintages.

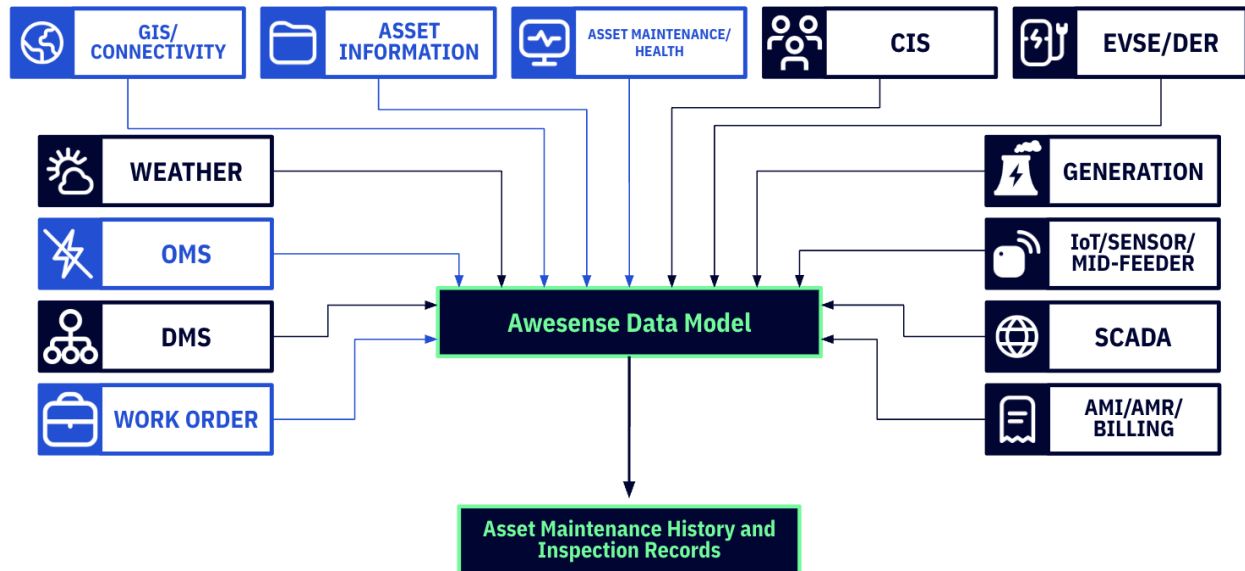
Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

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Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.